INCH-POUND
MIL-DTL-24643/63A(SH)
w/INT. AMENDMENT 1
11 July 2011

USED IN LIEU OF
MIL-DTL-24643/63A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSYSGU

This specification is approved for interim use by the Naval Sea Systems Command. Other activities in the Department of Defense may use this interim amendment or may continue using MIL-DTL-24643/63A.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable type is specifically designed for single circuit (see figure 1), three-phase, wye connected (with neutral wire), 60 or 400 Hertz, electrical power distribution. The designation Y represents the wye connection. Using two conductors per phase reduces the series inductance to 50 percent of the single conductor per phase value. In addition, the symmetrical spacing of the phase conductors from the neutral wire provides for balanced voltage drop when three-phase loads are applied.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Optional separator.
Third	-	Extruded silicone rubber, or silicone rubber-glass tape insulation at the manufacturer's option (see table I for diameter). Refer to table II for identification code.
Fourth	-	Seven conductors cabled together with a left-hand lay of six conductors around the center conductor (see figure 2 for cross-sectional view). The length of lay shall not exceed 24 times the pitch diameter. Fillers may be employed to obtain a well-rounded assembly.
Fifth	-	Optional binder.
Sixth	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. LSYSGU sizes and characteristics.

Military part no. M24643/63	Conductor						Cable		
	AWG	Circular MILS	Stranding	Amps per Phase (2 conductors)		Ω/1000 ft @ 25 °C (max.)	Jacket thickness (min.)	Overall diameter (max.)	Weight (lbs/ft)
				40 °C ambient	50 °C ambient				
-01UN	16	2580	7 x 0.0192	18	16	4.300	0.040	0.535	0.188
-02UN	14	4110	7 x 0.0242	29	27	2.680	0.040	0.580	0.255
-03UN	12	6530	7 x 0.0305	39	36	1.900	0.040	0.640	0.314
-04UN	10	10380	7 x 0.0385	62	58	1.125	0.040	0.730	0.420
-05UN	8	16510	7 x 0.0486	84	77	0.790	0.050	1.000	0.725
-06UN	4	41740	19 x 0.0469	145	133	0.287	0.050	1.206	1.400
-07UN	2	66360	37 x 0.0424	193	178	0.203	0.060	1.426	2.000
-08UN	1	83690	37 x 0.0476	237	218	0.145	0.060	1.541	2.400
-09UN	0	105600	61 x 0.0416	272	254	0.120	0.060	1.669	3.000
-10UN	00	133100	61 x 0.0467	320	294	0.0888	0.065	1.828	3.750
-11UN	000	167800	61 x 0.0524	376	346	0.0674	0.065	1.990	4.500
NOTE: All dimensions are in inches unless otherwise specified.									

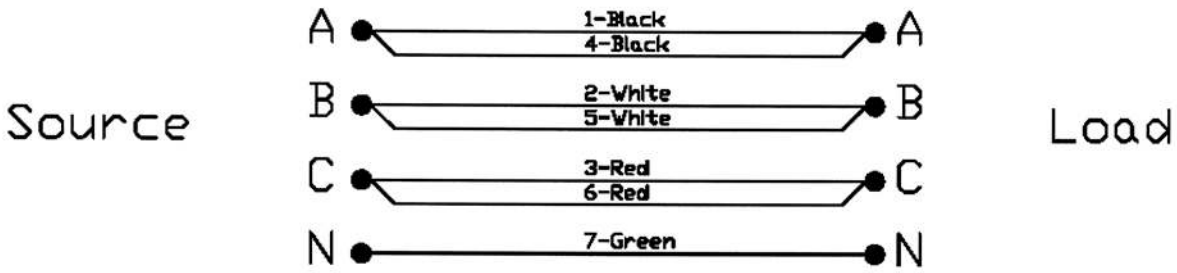


FIGURE 1. LSYSGU single circuit user diagram.

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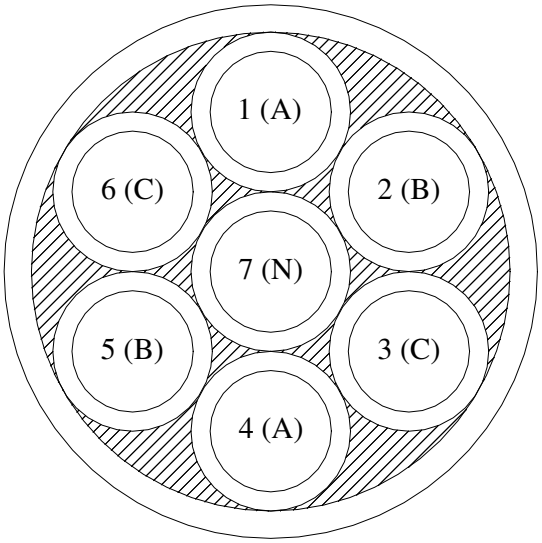


FIGURE 2. LSYSGU cross-section design.

TABLE II. LSYSGU identification code and color.

Conductor #	Identification code	Color
1	A	Black
2	B	White
3	C	Red
4	A	Black
5	B	White
6	C	Red
7	N	Green

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REQUIREMENTS:

Qualification Required.

INSPECTION:

Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (root mean square, min.)	
Sizes 16 through 10	3000 VAC
Sizes 8 and larger	5000 VAC
Insulation resistance (megohms/1000 feet, min.)	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Group B:	
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200±5 °C)	50
Tear (pounds per inch thickness, min.) (ASTM D470)	35
Gas flame (1 hour)	No failure
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	250
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Cold bending cable (mandrel is 12 times the outer diameter)	No damage
Group C:	
Physicals (aged)	
Jacket (cable) (air oven)	
Tensile strength (percent of unaged, min.)	60

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Elongation (percent of unaged, min.)	60
Hot oil immersion	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Shrinkage	No Failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	125
Group D:	
Flame propagation (cable)	No Failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Accelerated service (1.8 times the amps per phase @50 °C ambient)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Gas flame (3 hours)	No failure
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5

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Insulation	1.5
Durometer (jacket) – Type A (hardness, min.)	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSYSGU	500

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Preparing Activity:
Navy – SH
(Project 6145-2011-026)

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